Critical Path Analysis

Exam code: 9BS0

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Written by: Lisa Eades
Reviewed by: Steve Vorster
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The nature and purpose of critical path analysis

- Critical path analysis is a **project management tool** that uses **network analysis** to plan **complex** and **time-sensitive** projects
- Critical path analysis involves the construction of a visual model of the project that includes key elements:
  - A list of all **activities** required to complete the project
  - The time **(duration)** that each activity will take to complete
  - How each project activity **depends** on others
- Critical path analysis shows:
  - The order in which activities must be completed
  - The **longest path of project activities** to the completion of the project
  - The **earliest** and **latest** that each project activity can **start and finish** without delaying completion of the project as a whole
  - Activities within a project that can be carried out simultaneously
  - The **critical project activities** that, if delayed, will cause the project as a whole to overrun
  - Those project **activities where some delay is acceptable** without delaying the project as a whole
  - The **shortest time possible** to complete the project
- It allows managers to identify the **relationships between the activities** involved and to work out the **most efficient way of completing the project**
  - **Resources** such as raw materials and components can be ordered or hired at precisely the right time they are needed
  - **Working capital** may be managed efficiently
  - Where **delays** occur, managers can **identify the implications for the project's completion** and **redirect resources** if required

Stuck on something?

Ask questions and get help with exactly what you're revising.

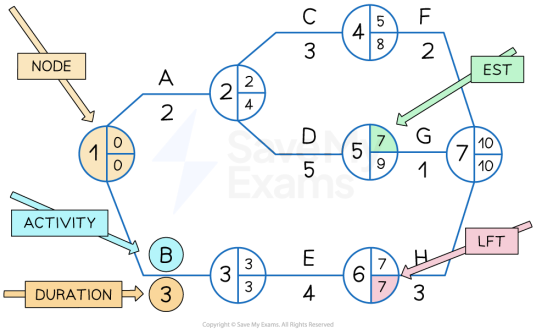
Got it

Drawing critical path analysis diagrams

Element	Description
Node	• A node is a **circle** that represents a point in time where an activity is started or finished • **The node is split into three sections** • The left half of the circle is the activity number • The top right section shows the earliest start time (EST) that an activity can begin based on the completion of the previous activity • The bottom right section shows the latest finish time (LFT) by which the previous activity must be completed
Activities	• An activity is a process or task within a project that takes time • Activities are shown on the network diagram as a **line that links nodes** • A description of the activity or a letter representing the activity is usually shown **above the line**
Duration	• The duration is the length of time it takes to complete an activity • The duration is shown as a **number of time units**, such as hours or days, **below the activity line**

Help me understand

Example network diagram

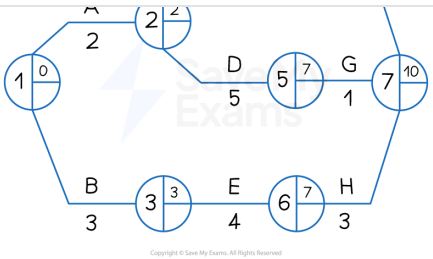


A simple network diagram showing key elements: node, activity and duration

- A network diagram must **always start and end at a single node**
- **Lines must not cross** and must **only be assigned to activities**

Critical path calculations

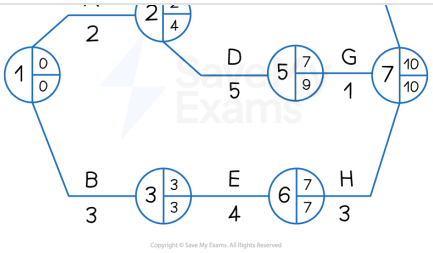
Earliest start times



An example of a simple network diagram showing earliest start times

- Working forwards from Node 1, it is possible to calculate the **earliest start time (EST)** for each activity by **adding the duration of each task**
- The EST for each activity is placed in the **top right** of each node
  - Node 1 is the starting point of the project and where both Activity A and Activity B begin
  - Activity A and Activity B are independent processes
  - Activity A has a duration of two days, and its EST is zero days
  - Activity B has a duration of three days, and its EST is also zero days
  - Activity C and Activity D both begin at Node 2 and are dependent upon the completion of Activity A but are independent from each other
    - Activity C has a duration of three days, and its EST is two days
    - Activity D has a duration of five days, and its EST is also two days
  - Activity E begins at Node 3
    - Activity E has a duration of four days, and its EST is three days
  - Activity F begins at Node 4
    - Activity F has a duration of two days, and its EST is five days
  - Activity G begins at Node 5
    - Activity G has a duration of one day, and its EST is seven days
  - Activity H begins at Node 6
    - Activity H has a duration of three days, and its EST is seven days
  - Node 7 is the endpoint of the project

Latest finish times



An example of a simple network diagram showing ESTs and LFTs

- Working backwards from Node 7, it is now possible to calculate the latest finish time (LFT) for each activity by **subtracting the duration of each task**
- The LFT for each activity is placed in the **bottom right** of each node
  - Node 7 is the endpoint of the project, which has an LFT of ten days
  - Activity H has a duration of three days
    - The LFT in Node 6 is seven days (10 days – 3 days)
  - Activity G has a duration of one day
    - The LFT in Node 5 is nine days (10 days – 1 day)
  - Activity F has a duration of two days
    - The LFT in Node 4 is eight days (10 days – 2 days)
  - Activity E has a duration of four days
    - The LFT in Node 3 is three days (7 days – 4 days)
  - Activity D has a duration of five days
    - The LFT in Node 2 is four days (9 days – 5 days)
  - Activity C has a duration of three days
    - The LFT in Node 3 is four days because Activity D is the more time-critical of the two activities that are dependent upon the completion of Activity A, so its LFT is recorded
  - Activity B has a duration of three days
    - The LFT in Node 1 is zero days (3 days – 3 days)
  - Activity A has a duration of two days
    - The LFT in Node 1 is zero days because Activity B is the more time-critical of the two starting activities, so its LFT is recorded
- The LFT in Node 1 is always zero

Identifying the critical path

- The critical path highlights those **activities that determine the length of the whole project**
  - If any of these critical activities are delayed, the **project as a whole will be delayed**
  - The critical path follows the **nodes where the EST and LFT are equal**
    - In the diagram below, Nodes 1, 3, 6 and 7 have equal ESTs and LFTs
    - Activities that determine these nodes are B, E and H

An example of a simple network diagram showing the critical path, BEH

Identifying and calculating float time

- Float time exists where there is a **difference between the EST and the LFT**
- Where float time is identified, managers may:
  - **Transfer resources**, such as staff or machinery, to more critical activities
  - **Allow extra time** to complete tasks to improve quality or allow for creativity

An example of a simple network diagram showing float nodes (4 and 5) and a critical node (6)

- The **total float** refers specifically to the spare time that is available so that the **overall project completion is not delayed**
- The **total float** for a specific activity is calculated using the formula

LFT for the activity – Duration of the activity – EST for the activity

- Using the diagram above, the following **total float times** can be calculated for Activities A to H

Activity	LFT	– Duration	– EST	= Total float

B	3	3	0	0
C	8	3	2	3
D	9	5	2	2
E	7	4	3	0
F	10	2	5	3
G	10	1	7	2
H	10	3	7	0

- The critical activities B, E and H each have a total float of zero days

Worked Example

The network diagram below shows the activities involved in a new promotional campaign for a small fashion accessories business, as well as the time (in weeks) it is expected that each activity will take to complete.

Calculate:

a) The ESTs and LFTs for each node [4]

b) The total float time for Activity G [3]

Step 1: Calculate the ESTs

Node 1 EST = **0**

Node 2 EST = 0 + 3 = 3, **but** 0 + 4 = 4 so **4**

Node 3 EST = 4 + 5 = **9**

Node 4 EST = 4 + 2 = **6**

Node 5 EST = 9 + 3 = **12**

Node 6 EST = 6 + 4 = **10**

[2]

Step 2: Calculate the LFTs

Node 8 = 15

Node 7 = 15 – 5 = 10

Node 6 = 15 – 4 = 11

Node 5 = 15 – 2 = 13

Node 4 = 11 – 4 = 7

Node 3 = 13 – 3 = 10

Node 2 = 10 – 6 = 4

Node 1 = 4 – 4 = 0 [2]

Step 3: Calculate the total float time for Activity G

$$Total\ float = LFT\ for\ the\ activity - Duration\ of\ the\ activity$$

$$= 11\ weeks - 4\ weeks - 6\ weeks$$

$$= 1\ week$$

[3]

Limitations of using critical path analysis

Limitations	Explanation
• Very lengthy or **complex projects involve a very large number of activities** that have numerous dependencies	• **Specialist network planning software** may be required • **Layers of supervision may be required** to manage different groups of activities within the project
• Network analysis often **relies on estimates and forecasts**	• Significant research is required **prior to the completion of network analysis** • Close and honest working relationships with suppliers are essential
• Network analysis **does not guarantee the success of a project**	• **Project managers** will need to be **highly skilled** and will need experience working with complicated plans

managers identify float periods	critical tasks • Machinery and other capital resources may require adaptation

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